THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **We Distribute, Yet Things Multiply** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 53	$x^2+y^2=(x+y)^2-2xy$
2	**(a)** 49	$x^2+y^2=(x-y)^2+2xy$
3	**(d)** 24000	$(512+488)(512-488)$
4	**(b)** 7	Square both sides
5	**(c)** 25	$p+q=a,\ pq=b$
6	**(d)** 141	25^2-22^2
7	**(b)** 47	$n^2-(n-1)^2=2n-1$
8	**(b)** $ab=0$	Compare with $a^2+2ab+b^2$
9	**(a)** 65	$x^2+y^2=(x+y)^2-2xy$
10	**(c)** 106	$x^2+y^2=(x-y)^2+2xy$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$x^2+y^2=(x+y)^2-2xy$ (1) $=144-70=74$ (1).	1+1
12	$(612+388)(612-388)=1000\times224$ (1) $=224000$ (1).	1+1
13	$p+q=8,\ pq=15$ (1); $p^2+q^2=8^2-2(15)=34$ (1).	1+1
14	$x^2+y^2=(x+y)^2-2xy$ (1) $=64-30=34$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x^2+y^2=(x-y)^2+2xy=16+42=58$ (1.5); $(x+y)^2=(x-y)^2+4xy=16+84=100$ (1.5).	1½+1½
16	$(3x+2)^2=9x^2+12x+4$ (1); $(3x-2)^2=9x^2-12x+4$ (1); difference $=24x$ (1).	1+1+1
17	$a^2+b^2=(10)^2-2(24)=52$ (1.5); $(a-b)^2=a^2+b^2-2ab=52-48=4$ (1.5).	1½+1½
	OR: $x^2+y^2=(x-y)^2+2xy=36+80=116$ (1.5); $(x+y)^2=(x-y)^2+4xy=36+160=196$ (1.5).	1½+1½
18	$(4x+5)^2=16x^2+40x+25$ (1); $(4x-5)^2=16x^2-40x+25$ (1); difference $=80x$ (1).	1+1+1
	OR: $a^2+b^2=(13)^2-2(40)=89$ (1.5); $(a-b)^2=a^2+b^2-2ab=89-80=9$ (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Area $=(3x+2)(3x-2)=9x^2-4$ (2); at $x=5$: 221 m² (1); square $=9x^2$, difference $=4$ m² (2).	2+1+2
	OR: $xy=\dfrac{(13)^2-97}{2}=36$ (2); $(x-y)^2=97-2(36)=25$ (2); $x-y=5$ (1).	2+2+1
20	Area $=(4x+3)(4x-3)=16x^2-9$ (2); at $x=6$: 567 m² (1); square $=16x^2$, difference $=9$ m² (2).	2+1+2
	OR: $xy=\dfrac{(16)^2-146}{2}=55$ (2); $(x-y)^2=146-2(55)=36$ (2); $x-y=6$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **We Distribute, Yet Things Multiply** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 508000	$(754+246)(754-246)$
2	**(c)** 14	Square both sides
3	**(a)** 41	$p+q=a$, $pq=b$
4	**(b)** 511	40^2-33^2
5	**(b)** 69	$n^2-(n-1)^2=2n-1$
6	**(c)** a^2+b^2	Expand
7	**(a)** 97	$x^2+y^2=(x+y)^2-2xy$
8	**(c)** 73	$x^2+y^2=(x-y)^2+2xy$
9	**(b)** 270000	$(635+365)(635-365)$
10	**(c)** 23	Square both sides

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$(845+155)(845-155)=1000\times690$ (1) $=690000$ (1).	1+1
12	$p+q=12$, $pq=32$ (1); $p^2+q^2=12^2-2(32)=80$ (1).	1+1
13	$x^2+y^2=(x+y)^2-2xy$ (1) $=196-66=130$ (1).	1+1
14	$(703+297)(703-297)=1000\times406$ (1) $=406000$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x^2+y^2=(x-y)^2+2xy=9+56=65$ (1.5); $(x+y)^2=(x-y)^2+4xy=9+112=121$ (1.5).	1½+1½
16	$(5x+3)^2=25x^2+30x+9$ (1); $(5x-3)^2=25x^2-30x+9$ (1); difference $=60x$ (1).	1+1+1
17	$a^2+b^2=(9)^2-2(18)=45$ (1.5); $(a-b)^2=a^2+b^2-2ab=45-36=9$ (1.5).	1½+1½
	OR: $x^2+y^2=(x-y)^2+2xy=25+72=97$ (1.5); $(x+y)^2=(x-y)^2+4xy=25+144=169$ (1.5).	1½+1½
18	$(2x+7)^2=4x^2+28x+49$ (1); $(2x-7)^2=4x^2-28x+49$ (1); difference $=56x$ (1).	1+1+1
	OR: $a^2+b^2=(15)^2-2(56)=113$ (1.5); $(a-b)^2=a^2+b^2-2ab=113-112=1$ (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Area $=(5x+4)(5x-4)=25x^2-16$ (2); at $x=2$: 84 m² (1); square $=25x^2$, difference $=16$ m² (2).	2+1+2
	OR: $xy=\dfrac{(11)^2-73}{2}=24$ (2); $(x-y)^2=73-2(24)=25$ (2); $x-y=5$ (1).	2+2+1
20	Area $=(2x+7)(2x-7)=4x^2-49$ (2); at $x=8$: 207 m² (1); square $=4x^2$, difference $=49$ m² (2).	2+1+2
	OR: $xy=\dfrac{(19)^2-193}{2}=84$ (2); $(x-y)^2=193-2(84)=25$ (2); $x-y=5$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **We Distribute, Yet Things Multiply** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 65	$p+q=a, pq=b$
2	**(d)** 272	36^2-32^2
3	**(c)** 95	$n^2-(n-1)^2=2n-1$
4	**(a)** $a=b$ or $a=-b$	$(a+b)(a-b)=0$
5	**(a)** 58	$x^2+y^2=(x+y)^2-2xy$
6	**(a)** 74	$x^2+y^2=(x-y)^2+2xy$
7	**(c)** 806000	$(903+97)(903-97)$
8	**(d)** 34	Square both sides
9	**(a)** 52	$p+q=a, pq=b$
10	**(d)** 768	52^2-44^2

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$p+q=15, pq=54$ (1); $p^2+q^2=15^2-2(54)=117$ (1).	1+1
12	$x^2+y^2=(x+y)^2-2xy$ (1) $=289-120=169$ (1).	1+1
13	$(960+40)(960-40)=1000\times920$ (1) $=920000$ (1).	1+1
14	$p+q=9, pq=14$ (1); $p^2+q^2=9^2-2(14)=53$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x^2+y^2=(x-y)^2+2xy=49+60=109$ (1.5); $(x+y)^2=(x-y)^2+4xy=49+120=169$ (1.5).	1½+1½
16	$(6x+1)^2=36x^2+12x+1$ (1); $(6x-1)^2=36x^2-12x+1$ (1); difference $=24x$ (1).	1+1+1
17	$a^2+b^2=(11)^2-2(30)=61$ (1.5); $(a-b)^2=a^2+b^2-2ab=61-60=1$ (1.5).	1½+1½
	OR: $x^2+y^2=(x-y)^2+2xy=64+96=160$ (1.5); $(x+y)^2=(x-y)^2+4xy=64+192=256$ (1.5).	1½+1½
18	$(3x+8)^2=9x^2+48x+64$ (1); $(3x-8)^2=9x^2-48x+64$ (1); difference $=96x$ (1).	1+1+1
	OR: $a^2+b^2=(14)^2-2(45)=106$ (1.5); $(a-b)^2=a^2+b^2-2ab=106-90=16$ (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Area $=(6x+5)(6x-5)=36x^2-25$ (2); at $x=3$: 299 m² (1); square $=36x^2$, difference $=25$ m² (2).	2+1+2
	OR: $xy=\dfrac{(16)^2-136}{2}=60$ (2); $(x-y)^2=136-2(60)=16$ (2); $x-y=4$ (1).	2+2+1
20	Area $=(7x+2)(7x-2)=49x^2-4$ (2); at $x=4$: 780 m² (1); square $=49x^2$, difference $=4$ m² (2).	2+1+2
	OR: $xy=\dfrac{(15)^2-173}{2}=26$ (2); $(x-y)^2=173-2(26)=121$ (2); $x-y=11$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.